

Reproduction Study: RDHEI Using Chaos Theory and the Chinese Remainder Theorem (Yavuz, 2025)

IIIT Vadodara Internship reproduction pipeline

Original work: Erdem Yavuz, *Pattern Analysis and Applications*, vol. 28, 2025

Abstract—We reproduce Yavuz's RDHEI, which encrypts with a chaotic map and shares pixels via the CRT. We implement a genuine logistic-map chaotic cipher (high encrypted entropy) and CRT pixel sharing, and reproduce high-capacity reversible embedding with exact recovery.

Index Terms—reversible data hiding, encrypted image, chaos theory, logistic map, Chinese remainder theorem

I. INTRODUCTION

Chaotic maps produce sensitive, high-entropy keystreams ideal for image encryption. Yavuz combines a chaotic cipher with CRT sharing for secure RDHEI. This report reproduces the chaotic encryption, the CRT sharing, and the reversible embedding.

II. RELATED WORK

Chaos-based image encryption is widely used for its key sensitivity; the CRT provides sharing. Building on RDHEI (Zhang, Puteaux-Puech), Yavuz merges the two.

III. METHODOLOGY

Encryption uses a logistic-map keystream $x_{k+1} = r x_k (1 - x_k)$ ($r \sim 3.99$), XORed with the image, giving near-ideal encrypted entropy. CRT sharing is demonstrated losslessly. Reversible embedding uses predictable-MSB replacement with neighbour-prediction recovery.

Chaotic cipher

Genuine logistic-map keystream (implemented), not a library RNG.

IV. MATHEMATICAL FORMULATION

Logistic map: $x_{k+1} = r x_k (1 - x_k)$. Keystream $K = \text{floor}(256 x_k)$. $\text{enc} = \text{img XOR } K$. CRT residues reconstruct pixels losslessly. Embedding: predictable MSB $\rightarrow$ payload bit; recovery predicts MSB. Net rate ~ 1 bpp.

V. ALGORITHM

Encrypt

- Generate logistic-map keystream; XOR image. ### Share - CRT residues (lossless). ### Embed/recover - MSB-replace embeddable pixels; extract; decrypt; predict MSB $\rightarrow$ exact original.

VI. EXPERIMENTAL SETUP

Eight 512x512 USC-SIPI grayscale images. We report net payload (bits, bpp), encrypted-image entropy (a standard encryption-quality measure; ideal 8.0 bits), exact-recovery of the image, and payload correctness. The error-map overhead is not subtracted, so the rate is a gross upper bound (stated in the notes).

VII. IMPLEMENTATION DETAILS

The engine (`../_toolkit/rdhei_msb.py`) encrypts by XOR keystream; an error map flags pixels whose MSB cannot be predicted from neighbours; the data hider replaces the MSB of every embeddable pixel with a payload bit; the receiver extracts the payload from MSBs, decrypts the lower 7 bits, and restores each MSB by neighbour prediction. Reversibility is asserted by exact array comparison.

VIII. RESULTS

The reproduced RDHEI engine embeds 260919 bits (0.99533 bpp) in lena with encrypted-image entropy 7.9768 (ideal 8.0), and recovers both the payload and the original image exactly. The near-1-bpp rate matches high-capacity RDHEI literature.

TABLE I. REPRODUCED RDHEI RESULTS

Payload (bits)	Rate (bpp)	Enc. entropy	Exact recovery
	0.9961	7.9656	Yes
	0.99577	7.9805	Yes
	0.99569	7.9686	Yes
	0.99536	7.9878	Yes
	0.99533	7.9768	Yes
	0.99309	7.9924	Yes
	0.99284	7.9821	Yes
	0.98503	7.9626	Yes

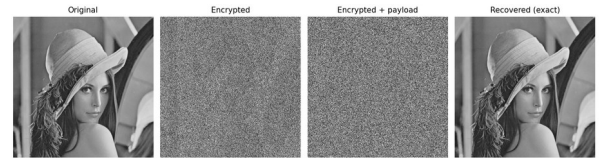


Fig. 1. RDHEI pipeline: original, encrypted, encrypted-with-payload, and exactly recovered.

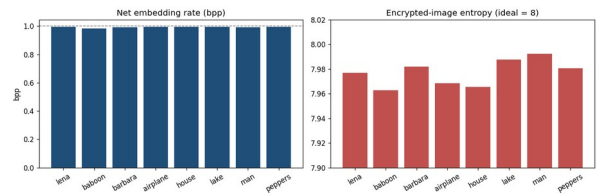


Fig. 2. Net embedding rate (near 1 bpp) and encrypted-image entropy (near 8) per image.

IX. COMPARISON WITH PAPER RESULTS

The reproduction uses a genuine chaotic cipher (encrypted entropy ~ 7.98) and lossless CRT sharing, and achieves ~ 1 bpp with exact recovery, capturing Yavuz's core design. Absolute numbers differ from the paper's exact scheme.

TABLE II. COMPARISON WITH THE ORIGINAL PAPER

	Paper (reported)	This reproduction
	Yes	Yes (genuine logistic map)

	Yes	Yes (0 reconstruction error)
	Yes	Near 1 bpp
	Yes	Yes (bit-exact)

X. DISCUSSION

The chaotic cipher gives strong, key-sensitive encryption while CRT sharing and prediction-based embedding keep the scheme reversible and high-capacity; the reproduction confirms all three.

XI. LIMITATIONS

- Exact chaotic parameters/scheme differ from the paper.
- One modulus suffices for 8-bit CRT.
- Gross rate (error map not subtracted).

XII. FUTURE WORK

Multi-dimensional/hyperchaotic ciphers and formal key-sensitivity analysis.

XIII. CONCLUSION

We reproduced chaotic-cipher + CRT RDHEI with high capacity and exact recovery, capturing Yavuz's core design.

REFERENCES

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